

Model Report: 3D CNN Patch-Level (5-seed ensemble)

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Script: /tmp/eval_3dcnn_field.py

Data Split:

train_count: 21741

test_count: 830

random_seed: 42

split_method: field-based (patch-level); metrics aggregated to FID

Metrics (test set):

Accuracy: 0.7506

Cohen Kappa: 0.6542

F1 Weighted: 0.7474

F1 Macro: 0.6997

Training Time: 6508s (108.5 min)

Hyperparameters:

target_size: [128, 128]

batch_size: 8

epochs: 30

optimizer: Adam(lr=1e-4)

loss: WeightedSparseFocalLoss(gamma=2.0, alpha=1/class_counts)

model_selection: val F1 macro (maximize)

early_stopping_patience: 6

data_pipeline: numpy patch store + vectorized reconstruct + tf.data parallel map/prefetch

conv3d_filters: [32, 64, 128]

bands: ['SA_B11', 'SA_B12', 'SA_B2', 'SA_B6', 'SA_EVI', 'SA_hue']

time_steps: 10

n_models: 5

seeds: [42, 101, 202, 303, 404]

reporting_level: field-level (majority vote by FID)

aggregation: mean softmax over 5 seeds (patch-level) -> argmax -> majority vote by FID

patch_level_accuracy: 0.7679

patch_level_kappa: 0.6977

per_seed_seconds: {'42': 1597.4, '101': 1821.5, '202': 1095.9, '303': 880.7, '404': 1098.4}

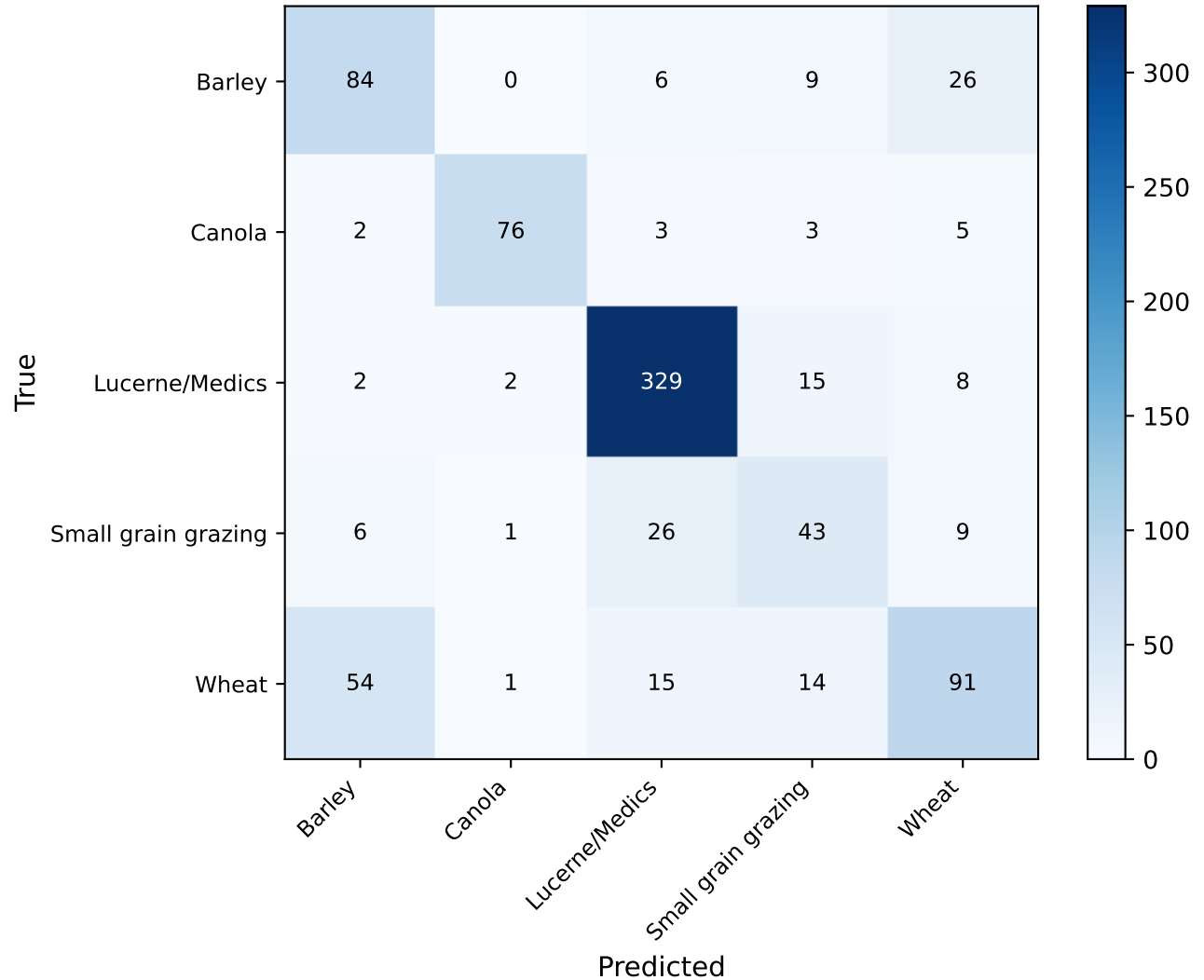
Notes:

Field-level metrics (majority vote by FID; 830 fields). Patch-level acc=0.7679, kappa=0.6977.

Per-Class Metrics

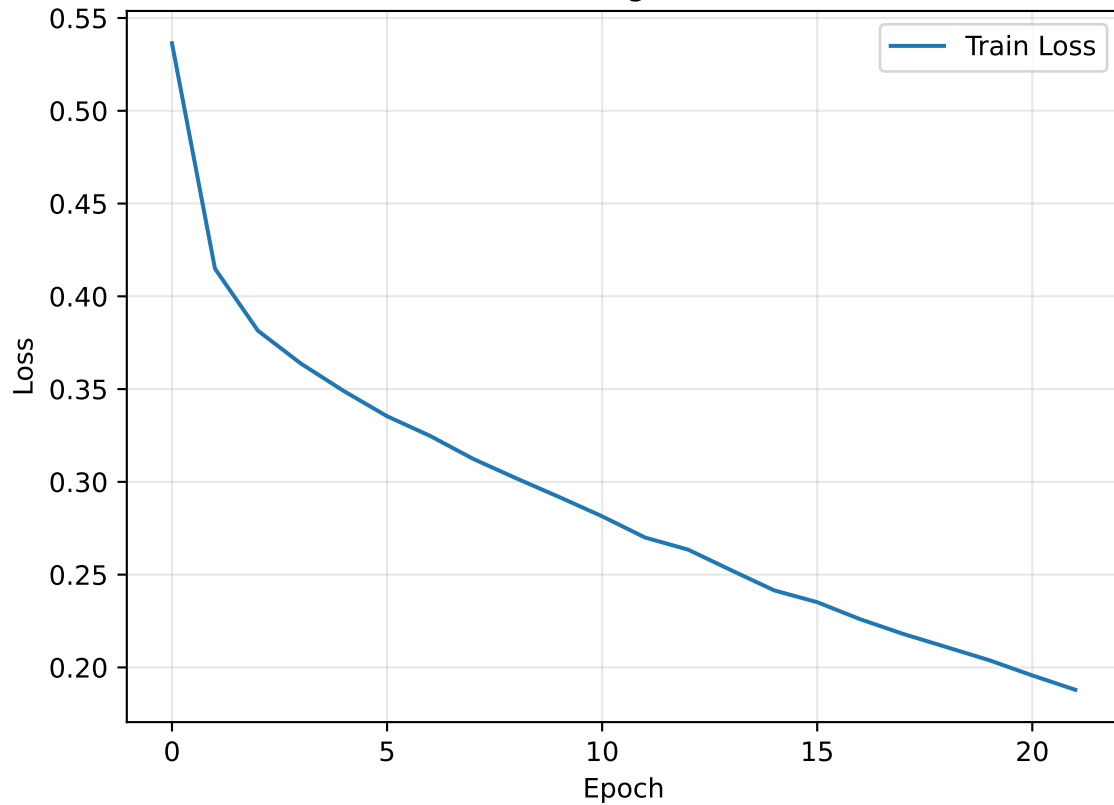
Class	Precision	Recall	F1-Score	Support
Barley	0.568	0.672	0.615	125
Canola	0.950	0.854	0.899	89
Lucerne/Medics	0.868	0.924	0.895	356
Small grain grazing	0.512	0.506	0.509	85
Wheat	0.655	0.520	0.580	175

Confusion Matrix



Training Curves

Training Loss



Training Accuracy

